Name: _____ ( ) Class: 26 / _____

ANDERSON SERANGOON JUNIOR COLLEGE

2026 JC2 Preliminary Examination

PHYSICS Higher 2 9478/02

Paper 2 Structured Questions **Friday 21 August 2026**

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, class index number and class in the spaces at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in the brackets [ ].

For Examiner's Use	
Paper 2 (75 marks)	
1	
2	
3	
4	
5	
6	
7	
8	
Deductions	
Total	

This document has **24** pages.

Data

speed of light in free space	$c = 3.00 \times 10^8$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}
	$(\frac{1}{4\pi\varepsilon_o} = 8.99 \times 10^9 \text{ m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{Gm}{r}$
temperature	$T / \text{K} = T/^\circ\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_o \sin \omega t$
velocity of particle in s.h.m.	$v = v_o \cos \omega t = \pm\omega\sqrt{x_o^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_o\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_o e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_o r}$
alternating current / voltage	$x = x_o \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_o I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_o N I}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_o n I$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_o e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 The Earth may be considered as a uniform sphere of radius 6.37 x 10^3 km.

Cambridge is at a point on the Earth's surface that has a latitude of 52.2° north of the Equator, as shown in Figure 1.1.

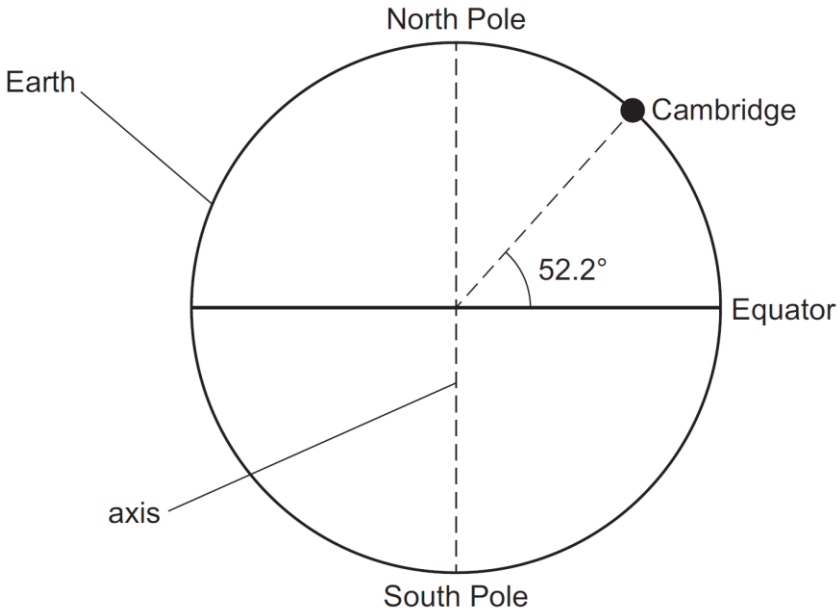


Figure 1.1

As the Earth spins on its axis, Cambridge moves in a circle that is parallel to the Equator but with a smaller radius.

(a) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

(i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = ... N [3]

(ii) On Figure 1.2, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in **1(a)(i)**.

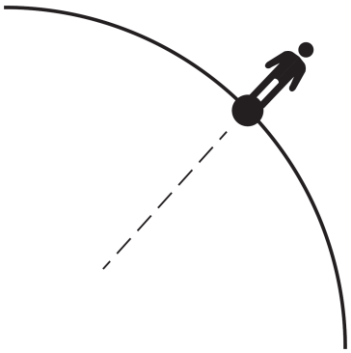


Figure 1.2

[2]

(b) Singapore has a small latitude north of Equator and can be assumed to be situated at the Equator. The Earth can be assumed to be a perfect sphere with uniform density.

State and explain how the apparent weight of another student of the same mass in Singapore compares to the student at Cambridge.

……..

……..

……..... [2]

[Total: 7]

2 **(a)** TESS is a satellite of mass 360 kg in a circular orbit about the Earth as shown in Figure 2.1. The satellite is launched from Earth's surface.

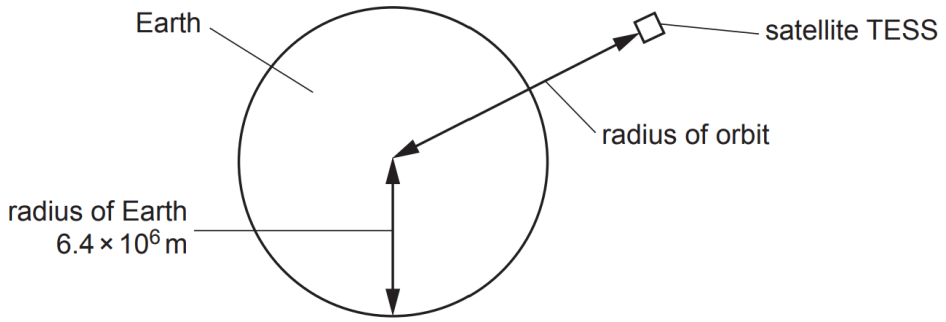


Figure 2.1 (not to scale)

The radius of the Earth is 6.4 x 10^6 m and the mass of the Earth, considered to be a point mass at its centre, is 6.0 x 10^{24} kg.

(i) It takes TESS 13.7 days to orbit the Earth.

Show that the radius of orbit of TESS is 2.4 x 10^8 m.

[3]

(ii) Calculate the change in gravitational potential energy between TESS in orbit and TESS on a launch pad on the surface of the Earth.

change in gravitational potential energy = ……..J [2]

(b) The Moon is Earth's only natural satellite, which may be considered to be a uniform sphere of diameter 3.4×10^3 km and mass 7.4×10^{22} kg. The Moon has no atmosphere.

During a collision of the Moon with a meteorite, a rock is thrown vertically up from the surface of the Moon with a speed of 2.8 km s^{-1}.

Assuming that the Moon is isolated in space, determine whether the rock will travel out into distant space or return to the Moon's surface.

[3]

[Total: 8]

3 **(a)** State the *first law of thermodynamics*, indicating the direction of all energy changes.

……

……

……[1]

(b) Use the first law of thermodynamics to explain why specific heat capacity of an ideal gas measured at constant volume is lower than the specific heat capacity when measured at constant pressure.

……

……

……

……

……

……[2]

(c) State two reasons why the temperature of a body is not a measure of the amount of internal energy in the body.

……

……

……

……

……

……[2]

(d) A car tyre has a fixed internal volume of 0.0360 m^3. The temperature and pressure of the air inside the car tyre is 25.0°C and 2.62 x 10^5 Pa initially. An additional 0.30 mol of air was pumped into the tyre at constant temperature of 25.0°C. Air can be considered as an ideal gas in this case.

(i) Show that the original amount of air particles in the car tyre is 3.8 mol.

[1]

(ii) Determine the final pressure of the gas inside the tyre.

pressure = …… Pa [2]

[Total: 8]

4 **(a)** Sound waves are longitudinal waves. State what is meant by a *longitudinal* wave.

.....

..... [1]

(b) A stationary sound wave in air has amplitude A. In an experiment, a detector is used to determine A^2. The variation of A^2 with distance x along the wave is shown in Figure 4.1.

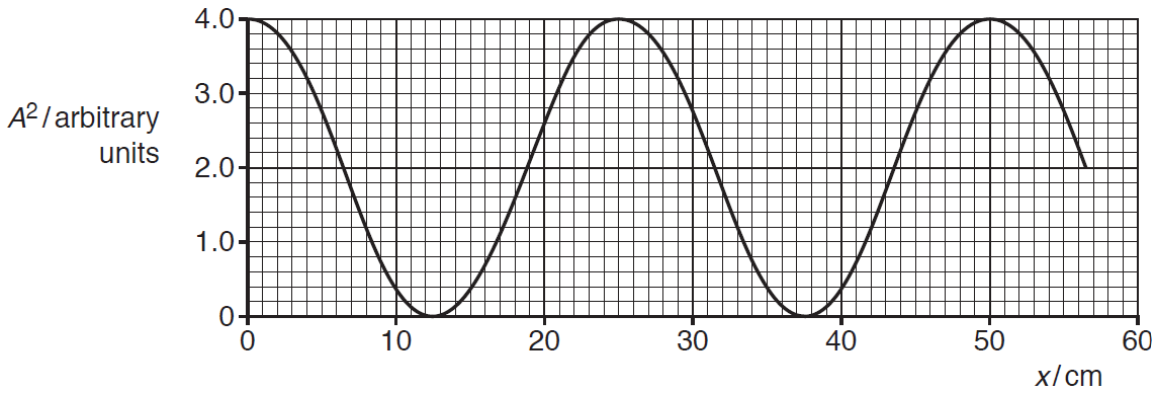


Figure 4.1

(i) State the phase difference between the vibrations of an air particle at x = 25 cm and the vibrations of an air particle at x = 50 cm.

phase difference = ... [1]

(ii) The speed of the sound in the air is 330 m s^{-1}. Determine the frequency of the sound wave.

frequency = ... Hz [2]

(iii) Determine the ratio

$$\frac{\text{intensity of wave at } x = 20\text{ cm}}{\text{intensity of wave at } x = 25\text{ cm}}$$

ratio = ... [2]

[Total: 6]

Turn over

5 **(a)** State the principle of superposition.

……

……

…… [2]

(b) An interference pattern is produced with the arrangement shown in Figure 5.1.

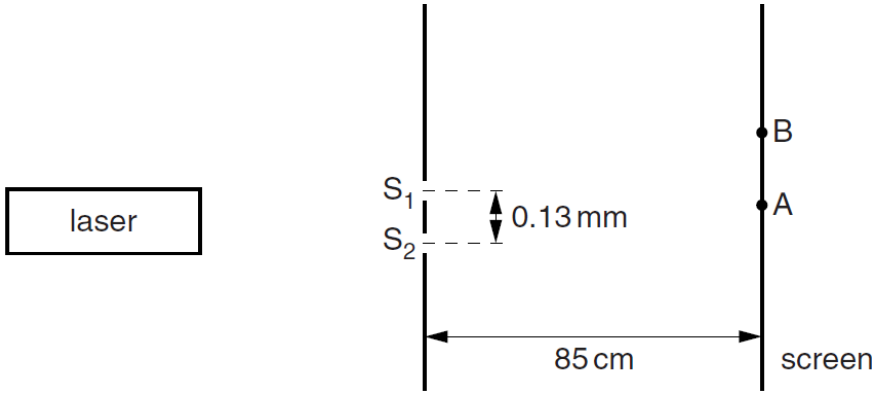


Figure 5.1 (not to scale)

Laser light of wavelength λ of 546 nm is incident on the slits S_1 and S_2. The slits are a distance 0.13 mm apart. The distance between the slits and the screen is 85 cm.

Two points on the screen are labelled A and B. The path difference between S_1A and S_2A is zero. The path difference between S_1B and S_2B is 2.5 λ. Maxima and minima of intensity of light are produced on the screen.

(i) Explain why a maxima is produced at point A.

……

……

……

…… [2]

(ii) Calculate the distance AB.

.

distance = ... m [3]

(iii) The laser is replaced by a laser emitting blue light. State and explain the change in the distance between the maxima observed on the screen.

……

…… [1]

(iv) The intensity of the laser passing through the two slits was initially the same. The intensity of the light through **one** of the slits is now reduced. Describe the changes, in any, in the appearance of the fringes on the screen.

……

……

……

…… [2]

[Total: 10]

6 **(a)** State the relationship between electric potential at a point and electric field strength at the point.

……

……

……

……[2]

(b) Two identical solid metal spheres A and B, each of radius 10 cm, are held in a vacuum such that the separation of their centres is 60 cm, as shown by Figure 6.1.

The charge on sphere A is +6.0 nC and the charge on sphere B is +3.0 nC.

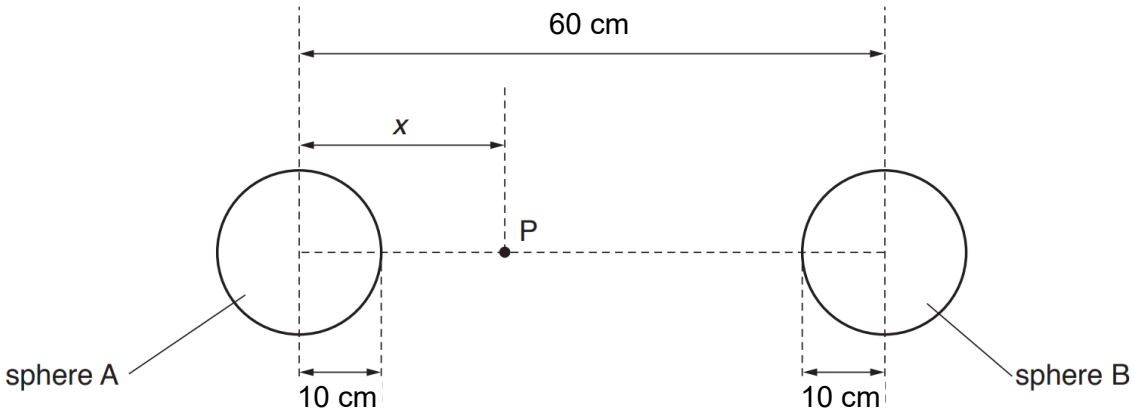


Figure 6.1

Assume that the charge on each sphere is a point charge at the centre of the sphere.

A movable point P is located on the line joining the centres of the two spheres.
The point P is a distance *x* from the centre of sphere A.

(i) On the axes of Figure 6.2, sketch a graph to show the variation with x of the electric potential V between the centres of the two spheres for x = 0 to x = 60 cm. Use the space above Figure 6.2 for any working.

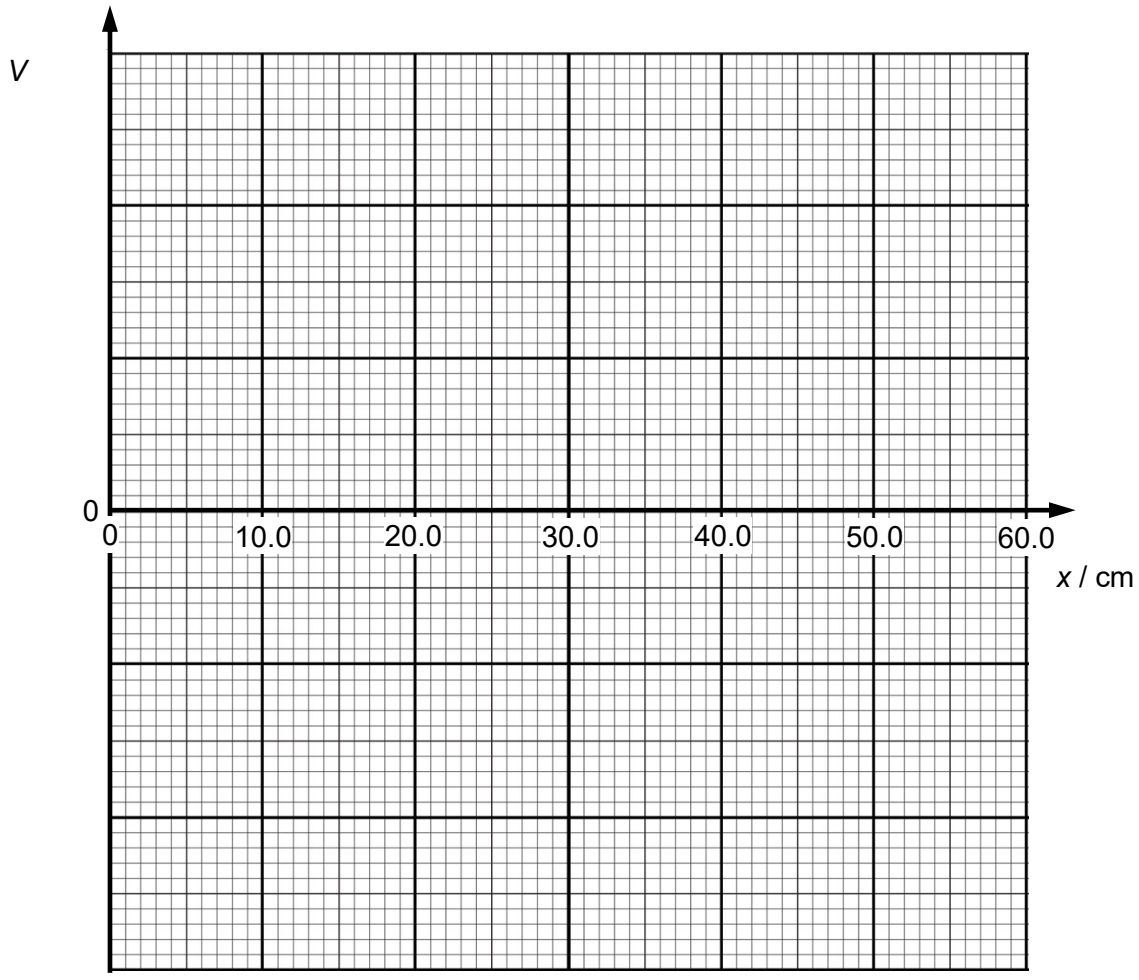


Figure 6.2

[4]

(ii) Calculate the electric potential energy E_P of the two spheres.

E_P = ……. J [2]

(iii) The two spheres are now released simultaneously so that they are free to move.

Describe and explain the subsequent motion of the spheres.

……

……

……

…… [2]

[Total: 10]

Turn over

7 **(a)** State what is meant by a *photon*.

……

……

……[2]

(b) Light in a beam has a continuous spectrum that lies within the visible region. The photons of light have energies ranging from 1.60 eV to 2.60 eV.

The beam passes through some hydrogen gas. It then passes through a diffraction grating and an absorption spectrum is observed.

(i) All of the light absorbed by the hydrogen is re-emitted. Explain why dark lines are still observed in the absorption spectrum.

……

……[1]

(ii) Some energy levels for the electron in an isolated hydrogen atom are illustrated in Figure 7.1.

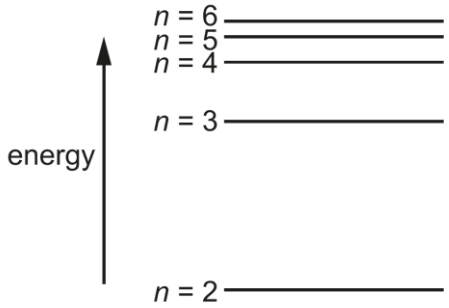


Figure 7.1

Table 7.1 shows the wavelengths of photons that are emitted in the transitions to $n = 2$ from the other energy levels shown in Figure 7.1. The energy associated with the energy level $n = 2$ is –3.40 eV.

Table 7.1

wavelength / nm
412
435
488
658

Calculate the energy, in J, of energy level $n = 3$.

energy = …… J [2]

(iii) Explain why the energy of each energy level is negative.

……

……[1]

[Total: 6]

8 Read the passage below and answer the questions that follow.

To bypass the biological limits of the human eyes in the night, engineers developed the Night Vision Goggles (NVGs) to allow soldiers and rescue workers to see in near-darkness. It converts very faint ambient light into a bright image through photoelectric effect.

Figure 8.1 shows a simplified schematic of an image-intensifier NVGs, comprising a photocathode, a microchannel plate (MCP), a phosphor screen and an eyepiece.

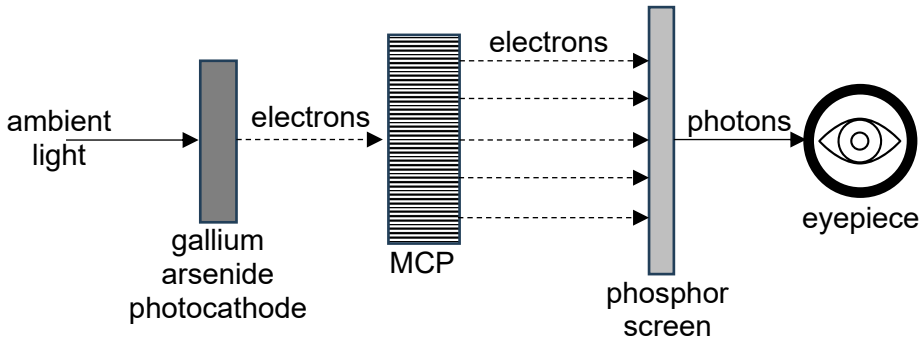


Figure 8.1

Ambient light from moonlight or starlight incidents onto a gallium arsenide (GaAs) photocathode in the NVGs, which has a work function of approximately 1.4 eV. The ejected photoelectrons are then accelerated through a potential difference of 150 V towards the MCP, a glass disc of thickness 480 μm drilled with millions of parallel channels, each of diameter 8.0 μm and at an angle of 13 ° to the normal of the MCP.

When a single electron enters a channel, it strikes the inner wall and releases secondary electrons via secondary emission. Each time an electron strikes the inner wall, 3 secondary electrons are emitted. These secondary electrons are themselves accelerated by a potential difference of 800 V applied across the MCP, striking the wall again and releasing further secondary electrons until a cascade of electrons exits the channel, as shown in Figure 8.2.

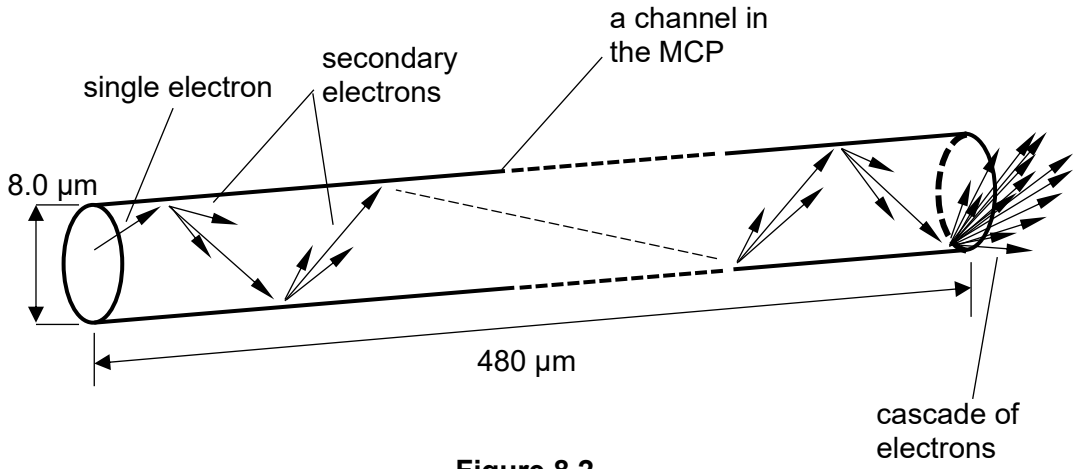


Figure 8.2

The cascade of electrons exit the MCP and is accelerated through a further potential difference of 4500 V across a short distance onto a phosphor screen. These accelerated electrons penetrate deep into the phosphor screen creating electron-hole pairs in the phosphor lattice. From these electron-hole pairs, photons are emitted from the phosphor screen. The emitted light is viewed through an eyepiece lens by the observer.

(a) The work function is defined as the minimum energy required to remove an electron from a metal surface.

(i) Determine the threshold wavelength of gallium arsenide.

threshold wavelength = …….. nm [2]

(ii) All objects above absolute zero emit radiation.

The peak wavelength of radiation λ_{peak} is governed by Wien's law:

$$\lambda_{peak}\, T = b$$

where T is the absolute temperature and b is the Wien's displacement constant which is approximately 2.898×10^{-3} m K.

Explain quantitatively whether the peak wavelength from a human body at a temperature of 37 °C can result in an emission of a photoelectron from gallium arsenide.

……..

……. [2]

(b) Calculate the speed of a photoelectron entering the MCP, assuming that it was emitted from the photocathode with negligible energy.

speed = …… m s^{-1} [2]

(c) The number of wall collisions n an electron undergoes in a channel of the MCP is given by

$$n = \frac{L}{2d} \sin \theta$$

where L is the length of the channel, d is the diameter of the channel and θ is the angle the axis the channel makes with the normal to the MCP surface.

(i) Suggest why the channels in the MCP are drilled tilted at an angle to the normal of the disc face.

……

……[1]

(ii) Determine the number of electrons exiting for each electron that enters a channel in the MCP.

number of electrons = …… [2]

(d) Suggest two reasons why the electrons exiting MCP are accelerated through a high potential difference of 4500 V across a short distance onto the phosphor screen.

1. ……..

……..

……..

2. ……..

……..

……..[2]

(e) Each high-energy electron accelerated towards the phosphor screen creates 450 electron-hole pairs in the phosphor lattice, emitting 350 photons of wavelength 545 nm.

(i) Determine the average energy required to create one electron-hole pair.

average energy = …… eV [2]

(ii) The internal quantum efficiency is given by the ratio of the total useful energy emitted as light to the energy of the incident electron.

Calculate the maximum internal quantum efficiency of the phosphor screen.

maximum internal quantum efficiency = …… [2]

(iii) Two spots formed on the phosphor screen subtend an angular separation of 3.3×10^{-4} rad at the observer's eye. In a dark environment, the dark-adapted pupil of the eye has a diameter of 7.0 mm.

Determine whether these two spots can be distinguished by the observer.

……..

…… [2]

(f) A student wanted to determine the value of Planck constant h by radiating gallium arsenide with monochromatic light of varying frequency f and measuring the stopping potential V_s needed to prevent photocurrent.

The equation relating the stopping potential V_s and the frequency of light f is given by

$$eV_s = hf - \phi$$

where e is the elementary charge and ϕ is the work function.

The variation of stopping potential V_s with frequency f obtained by the student is shown in Figure 8.3.

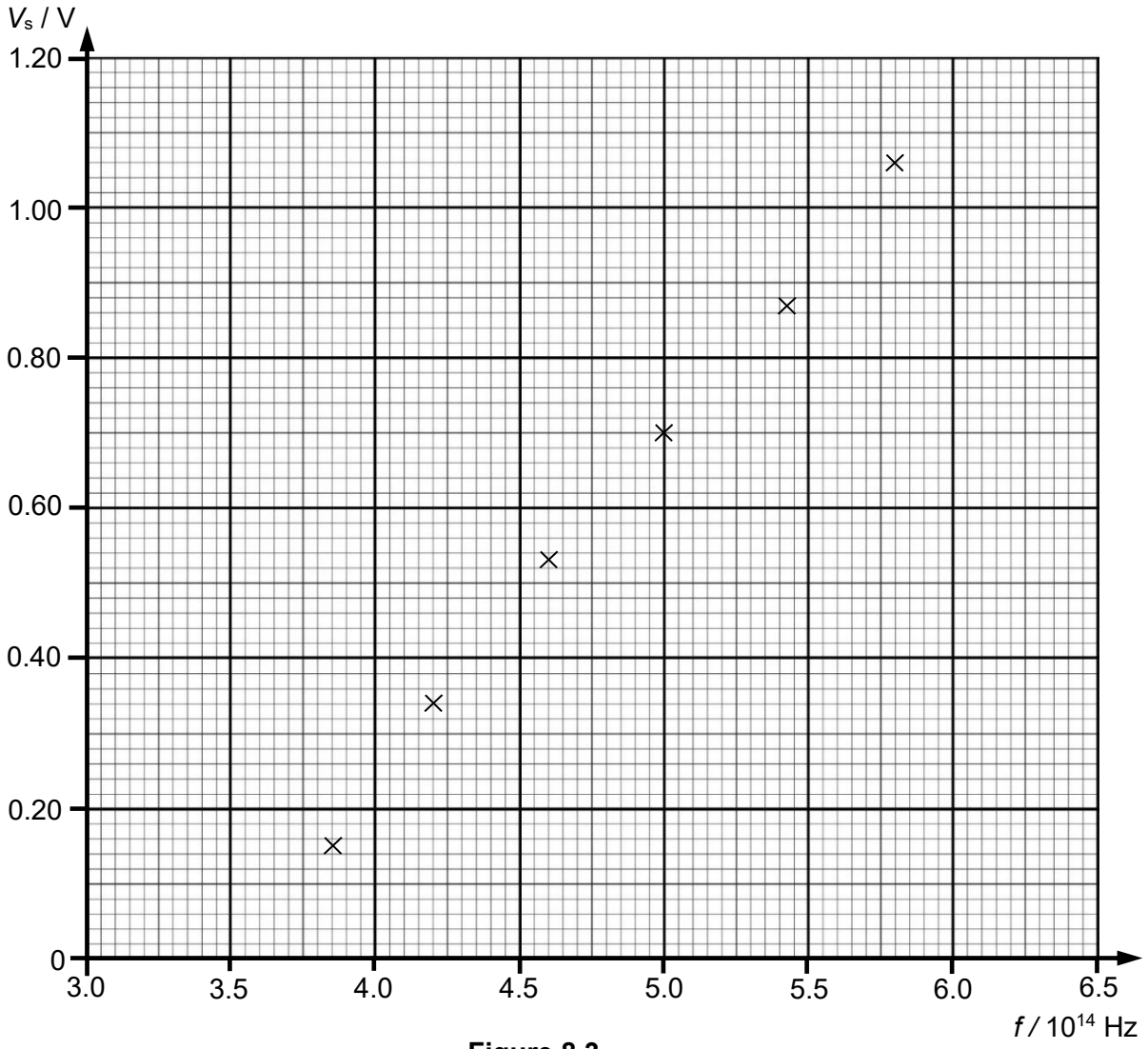


Figure 8.3

Use Figure 8.3 to determine the value of the Planck constant h.

h = …….. J s [3]

[Total: 20]